

## D-2 AI FDE Case Study 1

### Narrative

Risk/Fault Modes | Governance Controls | Evaluation Metrics | Stakeholder Tensions | Decision Rules

### CS01 - Airbus Skywise - Aviation Operations and Predictive Maintenance

#### Domain

#### Expanded narrative focus

|                                     |                                                                                                                                                                                                                                                                                                          |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aviation<br>Aerospace<br>Operations | / Maintenance, engineering, flight, reliability, and operations data were fragmented. The AI FDE problem is not only predicting aircraft component issues, but turning predictive signals into trusted, safety-aware maintenance decisions that reduce aircraft downtime without creating alert fatigue. |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Implementation context

A connected aviation data platform links in-flight, engineering, maintenance, reliability, and operations data. The workflow supports predictive alerting, fleet-performance analytics, aircraft health monitoring, reliability analysis, and maintenance decision support. Public case material reports significant operational impact, including the claim that Skywise was saving at least one aircraft out of service per day.

### Risk and failure modes

- False positives can ground aircraft unnecessarily, consume scarce maintenance capacity, and create planner distrust.
- False negatives can miss early signals and allow operational disruption or safety risk to grow.
- Older aircraft or inconsistent maintenance logs can reduce confidence and create biased coverage across fleets.
- Technically correct alerts can still be unusable if they arrive too late, lack parts/work-order context, or do not fit planner workflows.
- Engineering, maintenance, and flight operations may disagree because each optimizes for a different risk horizon.

### Governance controls

- Use an alert evidence card: component, aircraft, symptom, source systems, data recency, confidence, severity, similar historical cases, recommended action, alternatives, and accountable reviewer.
- Tier alerts by severity and confidence rather than using one global threshold.
- Embed recommendations into maintenance planning and work-order systems so planners do not need to copy insights manually.

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- Require human sign-off for grounding, urgent inspection, or safety-impacting decisions.
- Capture overrides, ignored alerts, completed actions, and post-event outcomes for continuous learning.
- Maintain data-lineage and data-quality flags, especially for older aircraft and inconsistent logs.

### Evaluation metrics

| Metric type                | Useful measures                                                                                                                        |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| AI quality                 | Aircraft out-of-service days avoided, Unscheduled maintenance reduction, Alert precision and recall by failure class and severity      |
| Workflow business value    | and Weighted false-positive and false-negative cost, Lead time between alert and required action, Planner acceptance and override rate |
| Adoption governance health | and Data coverage and data-quality score by fleet segment, Post-event learning closure rate                                            |

### Stakeholder tensions

- Flight operations wants aircraft availability and schedule protection.
- Engineering prioritizes safety and evidence strength.
- Maintenance teams care about capacity, parts, tooling, and inspection windows.
- Airline leadership wants measurable downtime reduction without increasing operational risk.

### Decision rules

- Recommend urgent escalation only when severity is high and evidence is sufficient or when severity is high enough to justify human triage despite weak evidence.
- Recommend inspection when evidence is medium/high and action is reversible or low-cost compared with risk.
- Recommend deferral when confidence is low, severity is low, and monitoring can safely continue.
- Never let model accuracy alone justify aircraft grounding; the decision must include operational severity, evidence, and accountable human approval.
- After every significant AI-influenced maintenance decision, run a post-event review comparing prediction, action taken, outcome, and learning update.

| Domain                                    | Expanded narrative focus                                                                                                                                                                                                                                     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Healthcare<br>Operations<br>Hospital Flow | Hospital operations require rapid coordination across beds, transfers, emergency department flow, operating rooms, staffing, and resources. The FDE problem is to improve patient flow without treating healthcare as a single-metric optimization exercise. |

### Implementation context

A command-center platform connects patient-flow, bed-availability, admissions, discharge, transfer, staffing, caregiver availability, and operational capacity data. It helps forecast demand and support bed planning, transfer decisions, and resource utilization. Reported outcomes include increased transfer volume and reduced ER hold time per ER admit.

### Risk and failure modes

- Reducing ED hold time may shift pressure to inpatient units, nurses, discharge teams, or post-acute pathways.
- An efficient bed assignment may be clinically inappropriate if patient acuity, infection-control needs, isolation, specialty care, or staffing skill mix is ignored.
- Operations staff may see too much clinical data or too little context to make safe operational decisions.
- Staff may ignore recommendations during peak load because recommendations are late, unrealistic, unexplained, or misaligned with local constraints.
- Low-confidence forecasts can create false certainty when operational pressure is high.

### Governance controls

- Define human override roles for bed managers, clinical leaders, charge nurses, and operations leaders.
- Use balancing metrics so the system cannot improve ED hold time by overloading another unit.
- Limit data visibility by role: operations teams see operational constraints; clinical details remain role-restricted unless needed.
- Log recommendation, rationale, confidence, user action, override reason, timestamp, and downstream impact.
- Establish escalation rules for clinically sensitive, capacity-critical, or low-confidence decisions.
- Review bottleneck shifts across the end-to-end patient journey, not only the command-center dashboard.

### Evaluation metrics

### Metric type

### Useful measures

AI quality

ED hold time and boarding time, Transfer acceptance and throughput, Occupancy by unit and acuity

Workflow and business value

Nurse workload and staffing variance, Time to bed assignment, OR delay/cancellation impact

Adoption governance health

and Override and acceptance rate, Patient-safety balancing indicators, Bottleneck shift indicators across units

### Stakeholder tensions

- Administrators want throughput and capacity utilization.
- Nurses and clinicians need safe staffing and clinically appropriate placements.
- Bed managers need fast, explainable, usable recommendations.
- Privacy and compliance teams require role-based access and defensible audit trails.

### Decision rules

- AI can recommend patient-flow actions, but clinical appropriateness and safety-sensitive exceptions require accountable human review.
- If one operational metric improves while downstream workload or safety worsens, the workflow must be redesigned before scaling.
- When forecast confidence is low, use scenario planning and human huddles instead of deterministic automated decisions.
- Expansion to another unit requires stakeholder sign-off, local workflow validation, balancing metrics, and audit-readiness.

## CS03 - AD Ports Group - AI Agents Workforce for Maritime Trade

### Domain

### Expanded narrative focus

Ports / Logistics  
/ Maritime  
Trade

Ports operate across vessels, terminals, containers, trucks, procurement, compliance, and daily operations. The FDE problem is scaling useful agents without creating uncontrolled agent sprawl, unsafe permissions, or conflicting operational advice.

### Implementation context

Agents are built for workflows such as vessel-speed optimization and container-utilization prediction using Azure AI Foundry and Copilot Studio. The operating model includes an AI Agents Directory and cross-functional participation from data science, engineering, cybersecurity, infrastructure, and domain users.

### Risk and failure modes

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- Leadership may chase agent count rather than operational value.
- Agents may receive excessive tool permissions, creating security and operational risk.
- Multiple agents may give conflicting recommendations for vessel speed, container allocation, or scheduling.
- Unofficial workarounds can bypass the agent directory if official agents are slow, untrusted, or hard to access.
- An agent with high usage but low value may continue consuming support effort because usage is mistaken for success.
- An agent action or recommendation can create safety, compliance, delivery, or financial impact.

### Governance controls

- Create an agent-intake gate: business owner, workflow problem, expected value, data sources, target users, permissions, risk tier, evaluation plan, and retirement criteria.
- Maintain an agent directory with owner, status, use case, risk tier, allowed tools, review date, telemetry, and support contact.
- Block write, delete, external communication, procurement commitment, schedule change, and compliance submission permissions by default unless approved.
- Use orchestration and conflict-resolution rules when multiple agents advise on the same operational decision.
- Require human approval for irreversible, high-cost, safety-impacting, customer-impacting, or regulatory actions.
- Define incident response: pause agent, preserve logs, notify owner, assess impact, remediate, and restart only after review.

### Evaluation metrics

#### Metric type

#### Useful measures

AI quality

Container-utilization improvement, Fuel and emissions impact, Vessel turnaround or schedule adherence

Workflow  
business value

and Agent task-completion rate, Tool-call success and failure rate, Human override and escalation rate

Adoption  
governance health

and Business outcome per agent, not agent count, Incidents, near misses, and blocked unsafe actions, Directory compliance and review timeliness

### Stakeholder tensions

- Executives may want speed and many agents.

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- Operations teams need reliability and workflow fit.
- Cybersecurity wants least privilege and containment.
- Sustainability teams may prefer fuel optimization that conflicts with delivery commitments.
- Domain users need recommendations that reflect real port constraints.

### Decision rules

- Build an agent only when the workflow has repeated decisions, measurable value, accessible data, and a clear accountable owner. Use simple automation when rules are deterministic; use a dashboard when human synthesis is the main need.
- Conflicting recommendations must be routed to a decision owner or orchestrator with assumptions, source data, and trade-off explanation.
- Retire or redesign an agent when business value is weak, telemetry is poor, risk is high, or users rely on workarounds.
- Sustainability recommendations must be balanced against safety, customer commitments, regulatory constraints, and operational SLAs.

## CS04 - Banco Bradesco - Multi-Agent Banking Platform

### Domain      Expanded narrative focus

|                                  |                                                                                                                                                                                                                                         |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Banking<br>Financial<br>Services | / Banking AI must support customer service and employee support without crossing regulated advice, fraud, KYC, privacy, security, or API-control boundaries. The FDE problem is safe orchestration, not simply a high containment rate. |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Implementation context

Bridge is a multi-agent GenAI platform using Azure OpenAI in Foundry Models and Azure API Management for deployment and scaling. The platform approach includes application governance, content safety, prompt management, agent intent classification, controlled API exposure, and escalation discipline. Reported outcomes include strong digital customer-service resolution, employee-query resolution, cost reduction, and faster product launches.

### Risk and failure modes

- A helpful answer may be interpreted as financial advice or product recommendation.
- Agents may route fraud, KYC, customer-service, employee-support, and document-processing journeys incorrectly.
- Unsafe write access to banking APIs can create unauthorized transactions, customer harm, or regulatory exposure.

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- Containment rate can become a bad metric if risky cases are kept inside automation instead of escalated.
- Agents can drift over time into unauthorized advice or unsafe responses.
- Weak logs make it difficult to explain disputed AI-assisted interactions to regulators.

### Governance controls

- Use intent classification and risk-tiered routing before answer generation or API access.
- Define advice boundaries: informational support is allowed; personalized financial advice, credit decisions, and regulated recommendations require approved workflow or human escalation.
- Classify APIs as read-only, draft-only, approval-required, or blocked.
- Log every route, intent, prompt, response, policy check, API call, tool result, escalation, human action, and final customer outcome.
- Evaluate do-not-answer behavior, refusal quality, escalation accuracy, content safety, groundedness, and API-call correctness.
- Monitor drift using sampled conversations, red-team prompts, complaint analysis, policy-violation tracking, and regulator-review simulations.

### Evaluation metrics

#### Metric type

#### Useful measures

AI quality

Safe resolution rate, Escalation precision and recall, Do-not-answer accuracy

Workflow and business value

Unauthorized-advice violation rate, API-call accuracy and blocked unsafe calls, Customer effort and satisfaction

Adoption and governance health

Complaint and dispute rate, Audit completeness, Cost per resolved interaction, Human-review turnaround time

### Stakeholder tensions

- Business wants fast resolution and cost reduction.
- Compliance wants safe escalation and advice boundaries.
- Customers want smooth service without unnecessary friction.
- Technology teams need reusable APIs and controlled scaling.
- Risk teams need audit evidence and defensible controls.

### Decision rules

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- When an interaction may be advice, fraud, KYC exception, complaint, financial hardship, credit decision, or customer harm scenario, route to specialist workflow or human review.
- A better metric than containment is safe resolution: correct answer, correct escalation, no policy violation, and auditable evidence.
- High-risk journeys should be automated only to draft, triage, or gather evidence unless regulatory and risk approvals permit action.
- Customer experience should remain smooth by explaining escalation as protection and continuity, not as a system failure.

### CS05 - Hughes / EchoStar - Production AI App Portfolio

#### Domain

#### Expanded narrative focus

|                                                                     |                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Satellite / Telecom /<br>Field Services /<br>Customer<br>Operations | The organization had productivity bottlenecks across sales-call auditing, customer-service summarization, retention, field-service automation, call analytics, and knowledge access. The FDE problem is moving from isolated tools to a reusable portfolio of production AI patterns. |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Implementation context

The portfolio uses Azure AI Foundry, Azure OpenAI Service, Azure AI Speech, RAG, speech-to-text, summarization, evaluation tools, and field-support automation. Public case material reports 12 production AI apps, expected annual work-hour savings, productivity boost, sales-call audit cost reduction, customer-service hours saved, and field-service hours saved.

#### Risk and failure modes

- Low-confidence transcripts can produce fluent summaries that appear more reliable than the source audio supports.
- Users may trust AI-generated summaries more than original calls, notes, or evidence.
- AI-generated summaries can become compliance or quality evidence without sufficient verification.
- A portfolio can create reusable leverage, but can also create maintenance sprawl if every app has custom components.
- High technical quality does not guarantee adoption or business value.
- ROI may be overstated when savings are distributed across departments and not tied to changed work behavior.

#### Governance controls

- Display transcript confidence, source links, speaker uncertainty, and missing-audio warnings in summaries.



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- Require mandatory human review for low-confidence transcripts, regulated claims, customer complaints, quality failures, disputes, refund/retention actions, and compliance evidence.
- Create reusable components: audio ingestion, speech-to-text, diarization, redaction, RAG, summarization, evaluation, logging, human review, and feedback capture.
- Prioritize apps using value, risk, feasibility, data readiness, reuse potential, and sponsor commitment.
- Measure behavior change, not only app launch count or usage volume.
- Define sunset criteria for weak adoption, weak value, high support cost, or inability to meet review thresholds.

### Evaluation metrics

| Metric type                    | Useful measures                                                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| AI quality                     | Speech recognition confidence and word-error rate, Summary factuality and source-grounding score, Human-audit agreement rate            |
| Workflow and business value    | and Mandatory-review trigger rate, Hours saved with baseline method, Cycle-time reduction                                               |
| Adoption and governance health | and Adoption depth by role and workflow, Reuse of common components, Cost per app and support load, Sunset/redesign decisions completed |

### Stakeholder tensions

- Sales and service teams want speed and automation.
- QA and compliance need defensible evidence.
- Field teams need practical, low-friction support.
- Technology teams need reusable architecture instead of one-off apps.
- Finance wants ROI evidence across distributed productivity gains.

### Decision rules

- If transcript confidence is low, the summary must show warnings, link to source segments, and require human review before operational or compliance use.
- Build apps first where value is measurable, data is ready, risk is controllable, and architecture components can be reused.
- Treat summaries as decision support unless validated and approved as evidence.
- Sunset a technically good app when adoption and behavior-change metrics remain weak after targeted redesign and enablement.

## CS06 - Moderna - Enterprise GPT Factory in Life Sciences

### Domain

### Expanded narrative focus

|               |                                                                      |
|---------------|----------------------------------------------------------------------|
| Life Sciences | / The organization scaled custom GPT creation across research,       |
| Biopharma     | / manufacturing, legal, commercial, and corporate functions. The FDE |
| Enterprise    | problem is balancing decentralized innovation with scientific rigor, |
| Productivity  | privacy, security, regulatory sensitivity, and operational control.  |

### Implementation context

ChatGPT Enterprise and earlier internal AI tooling enabled broad GPT creation. Public case material reports 750 GPTs, a high share of weekly active users creating GPTs, and high weekly usage. Use cases include clinical analysis support, contract summaries, policy bot, and investor-communication workflows. The challenge shifts from access to governance of distributed AI creation.

### Risk and failure modes

- Shadow AI and unmanaged GPTs may emerge outside approved platforms.
- Duplicate GPTs create inconsistent answers, duplicated effort, and unclear ownership.
- Informal GPTs can become business-critical without support, validation, or change control.
- Custom GPTs may mishandle sensitive life-sciences data, proprietary research, clinical information, legal material, manufacturing records, or regulatory content.
- Non-engineers building AI tools can create weak prompts, poor data boundaries, inadequate evaluation, and misleading confidence.
- Scientific uncertainty is different from ordinary business ambiguity; unsupported claims can harm research or regulatory decisions.
- Stale GPTs can continue producing outdated outputs after policy, process, study, product, or regulatory changes.

### Governance controls

- Maintain a GPT registry beyond name and owner: purpose, risk tier, data allowed, data blocked, user group, system instructions, knowledge sources, validation status, approval status, last review, usage telemetry, criticality, incident history, and retirement date.
- Risk-tier GPTs: personal productivity, team support, business workflow, regulated workflow, and scientific/regulatory decision support.
- Require review before use in research, legal, clinical, manufacturing, regulatory, external communication, or patient-impacting workflows.

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- Block or restrict sensitive data categories unless the platform, contract, access control, and purpose have been approved.
- Use AI champions, office hours, training, internal forums, and review boards to govern without killing innovation.
- Detect business-critical GPTs through usage concentration, dependency on recurring workflows, integration into deliverables, executive visibility, and lack of fallback process.
- Retire duplicate, unsafe, stale, ownerless, low-quality, or unreviewed GPTs using a registry-driven lifecycle process.

### Evaluation metrics

| Metric type                    | Useful measures                                                                                                                                                                        |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI quality                     | Validated productivity time saved by workflow, Output acceptance after human review, GPT registry completeness                                                                         |
| Workflow and business value    | High-risk GPT review completion, Duplicate GPT reduction, Incident and policy-violation rate                                                                                           |
| Adoption and governance health | Stale knowledge-source age, User adoption by meaningful workflow, not only conversation volume, Scientific/regulatory review pass rate, Critical GPTs with owners and fallback process |

### Stakeholder tensions

- Employees need freedom to build useful GPTs quickly.
- Scientific, legal, privacy, and regulatory teams need control over high-risk outputs.
- Executives want rapid adoption and measurable productivity.
- Security teams need data boundaries and monitoring.
- Domain experts need uncertainty and evidence handled rigorously.

### Decision rules

- Personal productivity GPTs may have lighter review if they do not process restricted data or influence regulated decisions.
- Regulated workflow GPTs require documented purpose, data controls, owner, validation, approval, monitoring, and retirement plan.
- Scientific outputs must show evidence, assumptions, uncertainty, source boundaries, and human expert review; they should not be presented as final scientific truth.
- High usage does not prove productivity; productivity must be tied to shorter cycle time, reduced rework, improved quality, or accepted deliverables.

- Duplicate, unsafe, stale, or low-quality GPTs should be consolidated, redesigned, blocked, or retired.

### Final Notes

- No generic AI answers. Strong answers must name workflow owner, risk trade-off, evidence needed, human accountability, control point, metric, and decision rule.
- Do not let participants optimize one metric blindly. Every case contains a balancing metric: safety, workload, compliance, adoption, data quality, or lifecycle risk.
- Treat reported outcome numbers carefully. Use them as case-study evidence, not as independently audited proof.
- Push from use case to operating model. The AI FDE bar is not the idea; it is the governed workflow that can survive production pressure.
- Use the scoring rubric consistently. A FDE-ready answer must connect discovery, architecture, governance, evaluation, business value, and stakeholder craft.

### Suggested participant instruction

Answer each probing question using this sequence: workflow problem -> decision owner -> risk/failure mode -> evidence required -> governance control -> evaluation metric -> decision rule. This prevents vague, tool-centric answers.